

BrandPulse

AI-Powered Social Media Monitoring Dashboard

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The Story: From Noise to Signal

Theme: From Noise to Signal

- ▷ **Past** — Brands manually scrolled Reddit, missing critical feedback and delaying responses.
- ▷ **Present** — BrandPulse automatically ingests, cleans, and analyses thousands of posts in real time and uses RSS for now.
- ▷ **Future** — A plug-and-play AI observatory for any online community.

Characters (The Team)

- ▷ **Ismail** — Platform & Data Engineer (ingestion, CI/CD, Docker, database design, frontend scaffolding)
- ▷ **Basim** — AI/ML Engineer (sentiment fine-tuning, zero-shot intent classifier) and frontend implementation
- ▷ **Momina** — Backend Engineer (FastAPI, REST APIs, database integration, backend testing)

The Problem Statement

Brands Struggle to Monitor Online Conversations

- ▷ Manual scanning of forums and social media is slow, inconsistent, and non-scalable.
- ▷ Critical issues, security alerts, billing complaints, technical bugs get buried in noise.
- ▷ No unified view of sentiment trends or intent breakdown across sources.

Consequence

- ▷ Missed opportunities to engage, delayed bug fixes, and security blind spots.

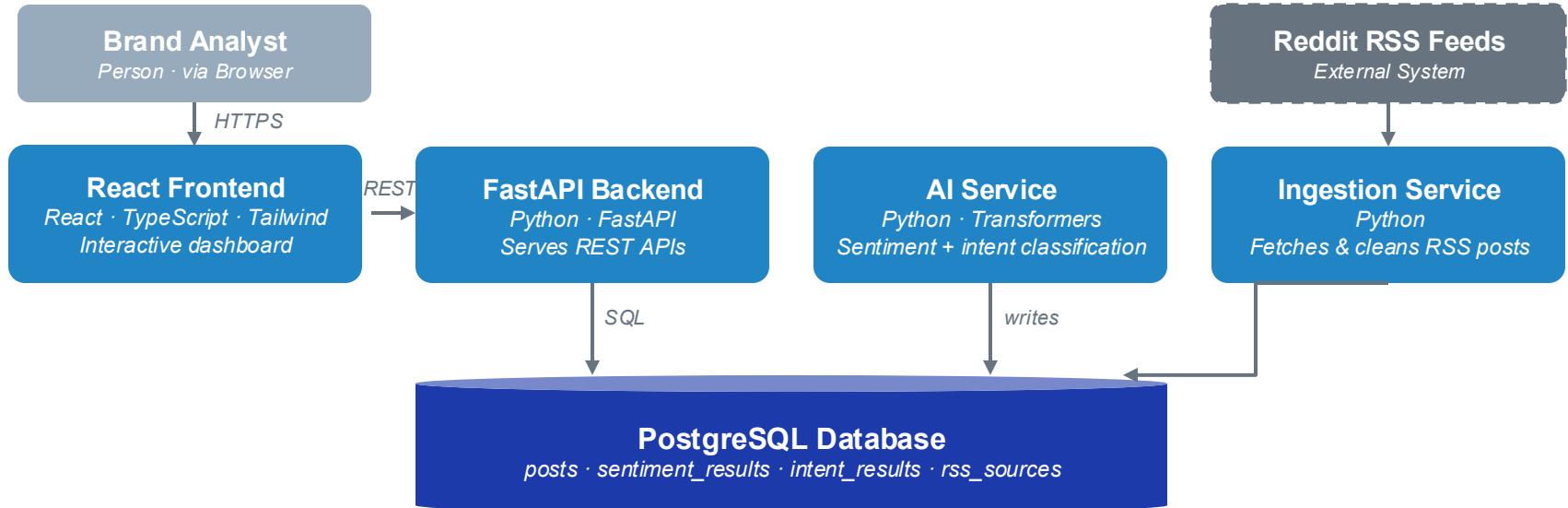
Solution Overview

BrandPulse — Automated, AI-Driven Social Listening

- ▶ **Ingestion Engine** — Pulls Reddit RSS feeds every 30 minutes, extensible to other sources.
- ▶ **AI Pipeline** — Fine-tuned sentiment model and zero-shot intent classifier categorise every post.
- ▶ **Live Dashboard** — Real-time KPI cards, charts, and a filterable post feed.
- ▶ **Scalable Architecture** — Dockerised microservices, ready for Azure.

Implementation Scope & Architecture

Containers



Scope: Reddit RSS only · 4 intent categories · 3 sentiment classes (positive / negative / neutral)

Work Division & Tools

Team Roles

- ▷ Ismail — repo setup, CI/CD, Docker Compose, RSS ingestion, DB models, React prototype
- ▷ Basim — sentiment model (DistilBERT), zero-shot intent classifier (DeBERTa-v3), AI pipeline design and frontend on React.
- ▷ Momina — FastAPI app, Pydantic schemas, REST endpoints, service layer, DB integration

Tech Stack

- ▷ Backend — Python, FastAPI, SQLAlchemy, Alembic, APScheduler
- ▷ AI — Hugging Face Transformers, PyTorch
- ▷ Database — PostgreSQL
- ▷ Frontend — React, TypeScript, Tailwind CSS, Recharts
- ▷ DevOps — Docker, Docker Compose, GitHub Actions, DVC, Azure (stretch)

Development Workflow & CI/CD

Trunk-Based Development with Feature Branches

- ▷ main ← develop ← feature/*
- ▷ Branch rules with required CI checks.

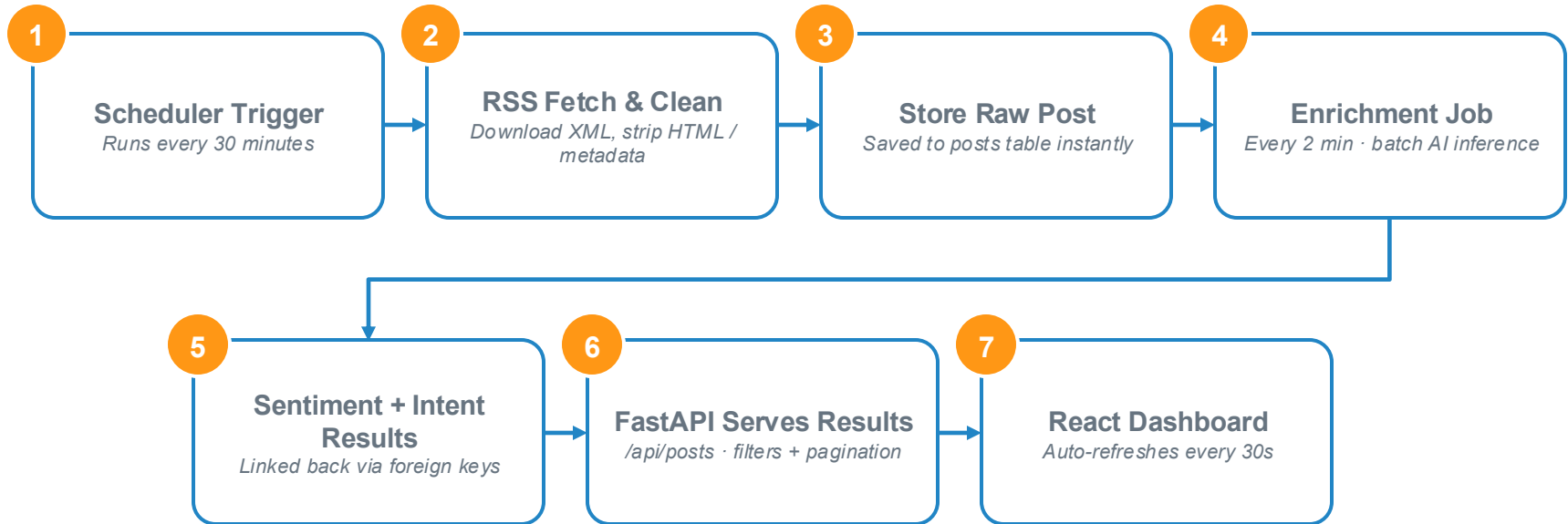
CI Pipeline (GitHub Actions)

- ▷ Runs on every pull request to develop and main.

Model Versioning

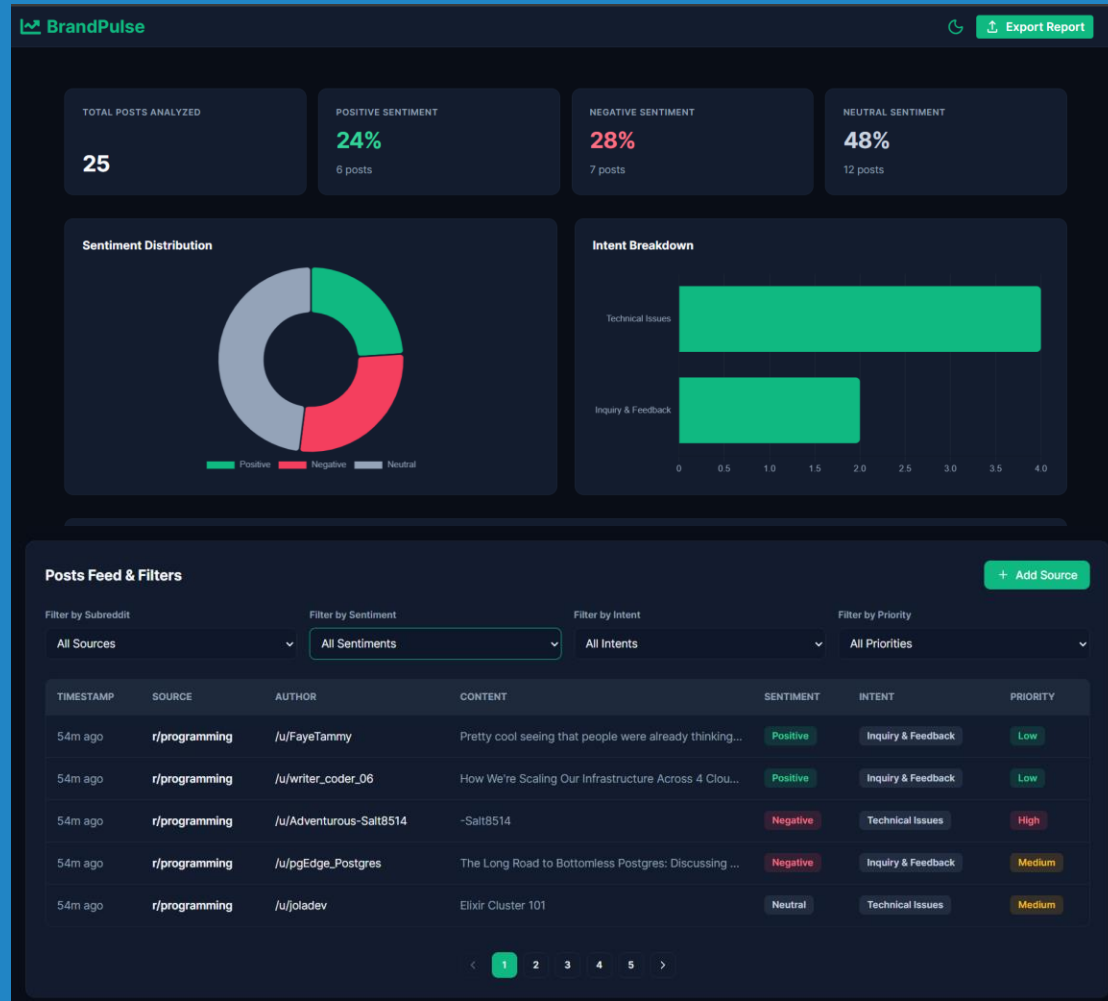
- ▷ DVC with a Google Drive remote — models are never committed to Git.

Data Flow (End-to-End)



Dashboard Demo

- KPI cards, sentiment pie, intent bar chart, and a timeline refreshed live
- Filterable post feed by source, sentiment, and intent, plus an Add Source modal



Key Achievements

- ▶ **End-to-End AI Pipeline:** Built a fully automated system that ingests Reddit RSS feeds, cleans text, runs dual AI models (sentiment + intent classification), and stores enriched results in a normalized PostgreSQL database.
- ▶ **Real-Time Dashboard Architecture:** Designed a two-stage "ingest first, enrich later" pattern using independent background schedulers, ensuring posts appear instantly while AI processing happens asynchronously.
- ▶ **Production-Grade Database Design:** Implemented a normalized schema with 17 optimized indexes across 5 tables, achieving sub-millisecond query performance even with joins and filters.
- ▶ **Professional DevOps Setup:** Established complete CI/CD with GitHub Actions, Docker containerization, DVC model versioning, and protected branching strategy from day one.

Future Directions

Short-Term (Post-Internship)

- ▷ Add OAuth-based Reddit API for higher rate limits.
- ▷ Implement batch AI inference for 10× enrichment speed.
- ▷ Deploy on Azure App Service with managed PostgreSQL.

Long-Term Vision

- ▷ Multi-source support (Twitter, news RSS, forums).
- ▷ Advanced analytics: trending topics, influencer tracking.
- ▷ User authentication and customizable alerts.

Thanks!

Any questions?

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GitHub: <https://github.com/Interns26/BrandPulse.git>